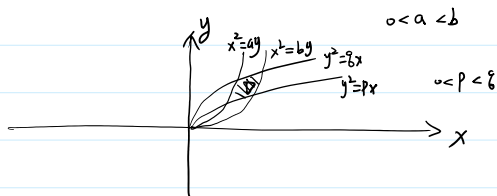


第15讲. 二重积分计算.

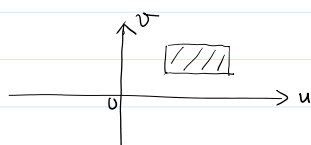
作业: P114. 1(1), (2), 2, 3, 5(3), (5), 6(1), (2), (4), 7(1), (3), 8(2), 9.

$$\iint_D f(x, y) dx dy \quad f(x, y) = 1.$$

$$\iint_D dx dy = S_D \quad \text{--- } D \text{ 面积.}$$



$$S_D = \iint_D dx dy \quad \begin{cases} u = \frac{x^2}{y} & a \leq u \leq b \\ v = \frac{y^2}{x} & p \leq v \leq q \end{cases}$$



$$dx dy = |\underline{J}| du dv$$

定理1. 设 D 为 oxy 平面上有界闭区域. Ω 为 $ou v$ 平面上有界闭区域.

$$\text{逆映射 } \varphi: \begin{cases} x = x(u, v) \\ y = y(u, v) \end{cases} \quad (u, v) \in \Omega \rightarrow D.$$

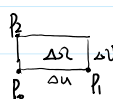
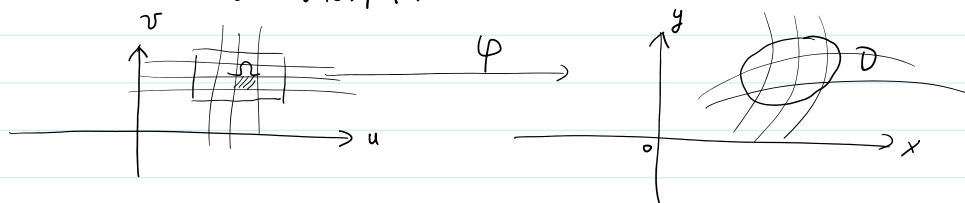
且 $\forall (u, v) \in \Omega$, Jacobian $\frac{\partial(x, y)}{\partial(u, v)} \Big|_{(u, v)}$ 可逆. 若 $f(x, y) \in C(D)$.

$$\text{则 } \iint_D f(x, y) dx dy = \iint_{\Omega} f(x(u, v), y(u, v)) \left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| du dv.$$

$$dx dy = \left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| du dv.$$

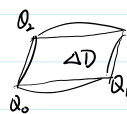
$$\frac{D(x, y)}{D(u, v)} = \det \frac{\partial(x, y)}{\partial(u, v)}$$

↓ Jacobian 行列式.



$$\Delta u > 0 \\ \Delta v > 0$$

$$\xrightarrow{\varphi}$$



$$P_0(u_0, v_0)$$

$$P_1(u_0 + \Delta u, v_0)$$

$$P_2(u_0, v_0 + \Delta v)$$

$$S_{\Delta D} \approx \|\vec{Q_0 Q_1} \times \vec{Q_0 Q_2}\|$$

$$Q_0 = \varphi(P_0)$$

$$Q_1 = \varphi(P_1)$$

$$Q_2 = \varphi(P_2)$$

$$\overrightarrow{Q_0 Q_1} = Q_1 - Q_0 = \varphi(P_1) - \varphi(P_0) = \frac{\partial(x, y)}{\partial(u, v)} \bigg|_{(u_0, v_0)} \begin{pmatrix} \Delta u \\ 0 \end{pmatrix} + \alpha(\Delta u)$$

$$\lim_{\Delta u \rightarrow 0} \frac{\|\alpha(\Delta u)\|}{\Delta u} = 0. \quad = \left(\frac{\frac{\partial x}{\partial u} \frac{\partial x}{\partial v}}{\frac{\partial y}{\partial u} \frac{\partial y}{\partial v}} \right) \bigg|_{(u_0, v_0)} \begin{pmatrix} \Delta u \\ 0 \end{pmatrix} + \alpha(\Delta u)$$

$$= \left(\frac{\frac{\partial x}{\partial u}}{\frac{\partial y}{\partial u}} \right) \bigg|_{(u_0, v_0)} \Delta u + \alpha(\Delta u)$$

$$\overrightarrow{Q_0 Q_2} = \varphi(P_2) - \varphi(P_0) = \frac{\partial(x, y)}{\partial(u, v)} \bigg|_{(u_0, v_0)} \begin{pmatrix} 0 \\ \Delta v \end{pmatrix} + \alpha(\Delta v)$$

$$\lim_{\Delta v \rightarrow 0} \frac{\|\alpha(\Delta v)\|}{\Delta v} = 0. \quad = \left(\frac{\frac{\partial x}{\partial v}}{\frac{\partial y}{\partial v}} \right) \bigg|_{(u_0, v_0)} \Delta v + \alpha(\Delta v)$$

$$S_{\Delta D} \approx \left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| \bigg|_{(u_0, v_0)} |\Delta u \Delta v| + o(\Delta u \Delta v) \quad \begin{pmatrix} \Delta u \rightarrow 0 \\ \Delta v \rightarrow 0 \end{pmatrix}$$

$$\downarrow$$

$$dx dy = \left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| \bigg|_{(u_0, v_0)} du dv.$$

$$S_D = \iint_D dx dy = \iint_{\Omega} \left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| du dv.$$

$$\forall T = \{\Delta D_1, \dots, \Delta D_n\}. \quad \forall (\xi_i, \eta_i) \in \Delta D_i. \quad \begin{cases} \xi_i = x(\bar{u}_i, \bar{v}_i) \\ \eta_i = y(\bar{u}_i, \bar{v}_i) \end{cases}$$

$$\downarrow$$

$$T' \quad \|T\| \rightarrow 0 \Rightarrow \|T'\| \rightarrow 0.$$

$$\iint_D f(x, y) dx dy = \lim_{\|T\| \rightarrow 0} \sum_{i=1}^n f(\xi_i, \eta_i) \Delta a_i$$

$$= \lim_{\|T\| \rightarrow 0} \sum_{i=1}^n f(\xi_i, \eta_i) \iint_{\Delta D_i} \left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$$

Δa_i 表示 ΔD_i 面积.

$$= \lim_{\|T\| \rightarrow 0} \sum_{i=1}^n f(\xi_i, \eta_i) \left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| \bigg|_{(u_i, v_i)} \Delta a_i$$

$$= \iint_{\Omega} f(x(u, v), y(u, v)) \left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| du dv.$$

例1. 计算 $I = \iint_{x^2+y^2 \leq 1} |3x+4y| dx dy$

解:
$$\begin{cases} u = \frac{1}{5}(3x+4y) \\ v = \frac{1}{5}(4x-3y) \end{cases} \quad \begin{pmatrix} \frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & -\frac{3}{5} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} u \\ v \end{pmatrix}$$

$$U$$

$$U^T U = U U^T = I.$$

$$x^2 + y^2 \leq 1.$$

$$u^2 + v^2 = (u, v) \begin{pmatrix} u \\ v \end{pmatrix} = (x, y) U^T U \begin{pmatrix} x \\ y \end{pmatrix} = x^2 + y^2 \leq 1.$$

$$I = \iint_{u^2+v^2 \leq 1} 5|u| \, du \, dv = \int_{-1}^1 5|u| \left(\int_{-\sqrt{1-u^2}}^{\sqrt{1-u^2}} dv \right) du = 10 \int_{-1}^1 |u| \sqrt{1-u^2} \, du$$

$$D_{uv} = \{ (u, v) \mid -1 \leq u \leq 1, -\sqrt{1-u^2} \leq v \leq \sqrt{1-u^2} \}$$

$$\stackrel{||}{=} 20 \int_0^1 u \sqrt{1-u^2} \, du$$

$$\stackrel{||}{=} 10 \int_0^1 \sqrt{1-u^2} \, d(1-u^2)$$

$$\stackrel{||}{=} \frac{20}{3}$$

例12. 计算 $I = \iint_D e^{\frac{x}{x+y}} \, dx \, dy$. $D = \{ (x, y) \mid x+y \leq 1, x \geq 0, y \geq 0 \}$.

解: $\begin{cases} u = x \\ v = x+y \end{cases} \quad \det \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} 1 & 0 \\ -1 & 1 \end{vmatrix} = 1.$

$$\downarrow \begin{cases} x = u \\ y = v-u \end{cases}$$

$$D_{uv} = \{ (u, v) \mid 0 \leq u \leq 1, u \leq v \leq 1 \}$$

$$I = \iint_{D_{uv}} e^{\frac{u}{v}} \, du \, dv = \int_0^1 \left(\int_u^1 e^{\frac{u}{v}} \, dv \right) du \quad \{ (u, v) \mid 0 \leq v \leq 1, 0 \leq u \leq v \}$$

$$= \int_0^1 \left(\int_0^v e^{\frac{u}{v}} \, du \right) dv = \int_0^1 v \left(e^{\frac{u}{v}} \Big|_{u=0}^{u=v} \right) dv$$

$$= \int_0^1 v(e-1) \, dv = \frac{1}{2}(e-1).$$

例13. 计算 $I = \iint_D \frac{(\sqrt{x} + \sqrt{y})^4}{x^2} \, dx \, dy$. $D: y=0, y=x, \sqrt{x} + \sqrt{y} = 1, \sqrt{x} + \sqrt{y} = 2$ (阴影).

解: $\begin{cases} u = \sqrt{x} + \sqrt{y} \\ v = \frac{y}{x} \end{cases} \quad 1 \leq u \leq 2, \quad 0 \leq v \leq 1.$

$$dx \, dy = \left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| \, du \, dv = \frac{2x^2}{u} \, du \, dv$$

$$I = \iint_{[1,2] \times [0,1]} \frac{u^4}{x^2} \cdot \frac{2x^2}{u} \, du \, dv$$

$$\stackrel{||}{=} \frac{1}{\left| \det \frac{\partial(u, v)}{\partial(x, y)} \right|} = \frac{1}{\left| \begin{vmatrix} \frac{1}{2\sqrt{x}} & \frac{1}{2\sqrt{y}} \\ -\frac{y}{x^2} & \frac{1}{x} \end{vmatrix} \right|}$$

$$= \iint_{[1,2] \times [0,1]} 2u^3 \, du \, dv$$

$$\stackrel{||}{=} \frac{1}{\frac{1}{2x^{\frac{3}{2}}} + \frac{1}{2x^{\frac{3}{2}}} \cdot \sqrt{v}} = \frac{\sqrt{x} + \sqrt{y}}{2x^2}$$

$$= \int_1^2 2u^3 \, du \cdot \int_0^1 dv = \frac{15}{2}.$$

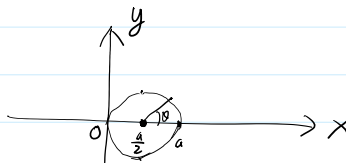
$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

$$\det \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r$$

$$\iint_D f(x, y) dx dy = \iint_{D_{\theta r}} f(r \cos \theta, r \sin \theta) \underline{r dr d\theta}.$$

例4. 设 $a > 0$. 求 $I = \iint_{x^2+y^2 \leq ax} xy^2 dx dy$.

解:
$$\begin{cases} x = \frac{a}{2} + r \cos \theta \\ y = r \sin \theta \end{cases} \quad \begin{aligned} \theta &\in [0, 2\pi] \\ r &\in [0, \frac{a}{2}] \end{aligned}$$



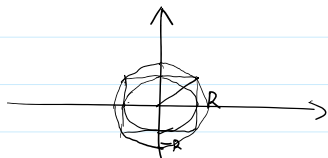
$$I = \int_0^{2\pi} d\theta \int_0^{\frac{a}{2}} (\frac{a}{2} + r \cos \theta) r^2 \sin^2 \theta \cdot r dr$$

$$= \int_0^{2\pi} \left(\int_0^{\frac{a}{2}} r^3 \sin^2 \theta dr \right) d\theta + \int_0^{2\pi} \left(\int_0^{\frac{a}{2}} r^4 \cos \theta \sin^2 \theta dr \right) d\theta$$

$$= \frac{a^5}{128}$$

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \quad \begin{aligned} \theta &\in [-\frac{\pi}{2}, \frac{\pi}{2}] \\ r &\in [0, a \cos \theta] \end{aligned}$$

例5. 求 $I = \int_{-\infty}^{+\infty} e^{-x^2} dx = \lim_{R \rightarrow +\infty} \underbrace{\int_{-R}^R e^{-x^2} dx}_{I_R}$



$$\begin{aligned} I_R^2 &= \int_{-R}^R e^{-x^2} dx \cdot \int_{-R}^R e^{-y^2} dy = \int_{-R}^R \int_{-R}^R e^{-(x^2+y^2)} dx dy \\ &= \iint_D e^{-(x^2+y^2)} dx dy \end{aligned}$$

$$D = [-R, R] \times [-R, R].$$

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

$$D_1 = \{(x, y) \mid x^2 + y^2 \leq R^2\},$$

$$D_1 \subset D \subset D_2$$

$$D_2 = \{(x, y) \mid x^2 + y^2 \leq 2R^2\}.$$